Gas Laws

21. To double the volume of a given mass of an ideal gas at 27°*C* keeping the pressure constant, one must raise the temperature in degree centigrade to

(a) 54° (b) 270°

(c) 327° (d) 600°

22. Which of the following statements about kinetic theory of gases is wrong

(a) The molecules of a gas are in continuous random motion

(b) The molecules continuously undergo inelastic collisions

(c) The molecules do not interact with each other except during collisions

(d) The collisions amongst the molecules are of short duration

23. If a Vander-Waal's gas expands freely, then final temperature is

(a) Less than the initial temperature

(b) Equal to the initial temperature

(c) More than the initial temperature

(d) Less or more than the initial temperature depending on the nature of the gas

24. In Vander Waal's equation $\left[P + \frac{a}{V^2}\right](V - b) = RT$, the dimensions of *a* are

(a) $M^1L^5T^{-2}$ (b) $M^0L^2T^{-3}$

(c) $M^1L^3T^{-2}$ (d) $M^1L^1T^{-2}$

25. At NTP, sample of equal volume of chlorine and oxygen is taken. Now ratio of No. of molecules

(a) 1 : 1 (b) 32 : 27

(c) 2 : 1 (d) 16 : 14

26. A gas at the temperature 250 *K* is contained in a closed vessel. If the gas is heated through 1*K*, then the percentage increase in its pressure will be

(a) 0.4% (b) 0.2%

(c) 0.1% (d) 0.8%

27. A cylinder of 5 *litre* capacity, filled with air at N.T.P. is connected with another evacuated cylinder of 30

litres of capacity. The resultant air pressure in both the cylinders will be

(a) 38.85 *cm* of Hg

(b) 21.85 *cm* of Hg

(c) 10.85 *cm* of Hg

(d) 14.85 *cm* of Hg

28. Volume of gas become four times if

(a) Temperature become four times at constant pressure

(b) Temperature become one fourth at constant pressure

(c) Temperature becomes two times at constant pressure

(d) Temperature becomes half at constant pressure

29. The relationship between pressure and the density of a gas expressed by Boyle's law, $P = KD$ holds true

(a) For any gas under any conditions

(b) For some gases under any conditions

(c) Only if the temperature is kept constant

(d) Only if the density is constant

30. Kinetic theory of gases provide a base for

(a) Charle's law

(b) Boyle's law

(c) Charle's law and Boyle's law

(d) None of these

31. On $0°C$ pressure measured by barometer is 760 *mm*. What will be pressure on $100°C$

(a) 760 *mm*

(b) 730 *mm*

(c) 780 *mm*

(d) None of these

32. A perfect gas at $27°C$ is heated at constant pressure to $327°C$. If original volume of gas (at $27°C$ is V then volume at $327°C$ is

(a) V (b) $3V$

(c) $2V$ (d) $V/2$

33. Hydrogen gas is filled in a balloon at $20°C$. If temperature is made $40°C$, pressure remaining same, what fraction of Hydrogen will come out

(a) 0.07 (b) 0.25

(c) 0.5 (d) 0.75

34. At constant temperature on increasing the pressure of a gas by 5% will decrease its volume by

(a) 5% (b) 5.26%

(c) 4.26% (d) 4.76%

35. Which law states that effect of pressure is same for all portion

(a) Pascal's law

(b) Gallusac's law

(c) Dalton's law

(d) None of these

36. When the pressure on 1200 *ml* of a gas in increased from 70 *cm* to 120 *cm* of mercury at constant temperature, the new volume of the gas will be

(a) 700 *ml* (b) 600 *ml*

(c) 500 *ml* (d) 400 *ml*

37. The equation for an ideal gas is $PV = RT$, where V represents the volume of

(a) 1 *gm* gas

(b) Any mass of the gas

(c) One *gm mol* gas

(d) One *litre* gas

38. The molecular weights of O_2 and N_2 are 32 and 28 respectively. At 15°*C*, the pressure of 1 *gm* O_2 will be the same as that of 1 *gm* N_2 in the same bottle at the temperature

(a) – 21°*C* (b) 13°*C*

(c) 15°*C* (d) 56.4°*C*

39. At 100 *K* and 0.1 atmospheric pressure, the volume of helium gas is 10 *litres*. If volume and pressure are doubled, its temperature will change to

(a) 400 *K* (b) 127 *K*

(c) 200 *K* (d) 25 *K*

40. Under which of the following conditions is the law $PV = RT$ obeyed most closely by a real gas

(a) High pressure and high temperature

(b) Low pressure and low temperature

(c) Low pressure and high temperature

(d) High pressure and low temperature

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